

TELANGANA STATE-2015
CLASS 10
GENERAL SCIENCE, Paper – II

Time: 2 hours 45 min.]

[Maximum Marks: 40

Instructions:

- (i) The Question paper comprises of FOUR Sections - I, II, III and IV.
- (ii) All the Questions are compulsory.
- (iii) There is no overall choice. However there is internal choice to the Questions under Section-I.
- (iv) In the time duration of 2 hrs. 45 minutes, 15 minutes of time is exclusively allotted to read and understand the Question paper.

SECTION -I

(4x4=16 marks)

Note:

- (i) Answer all the questions.
- (ii) Each question carries FOUR marks.
- (iii) There is internal choice for each question. Only one option from each question is to be attempted.

(iv) Answer each question in 8 to 10 sentences.

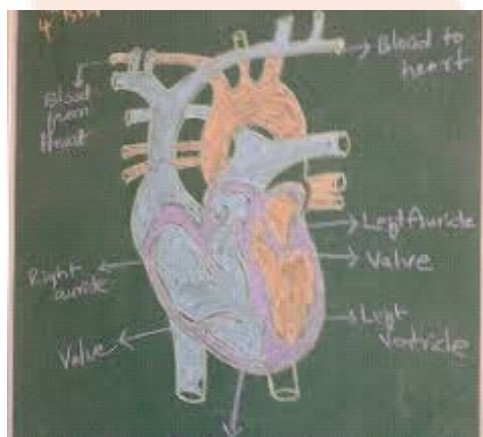
1. How can you say that Kidney is suitable for filtration of biological waste from blood in man?

[Ans] All the biological waste or toxic waste is excreted with the help of kidney. It is known as urine and it basically filter about 750-1000 pints of blood. The kidney filters it and circulate the same to rest of the body. Waste is then sent to bladder and then the same waste is out of the body in the form of urine.

OR

What is called pumping station in human body? Explain its structure with suitable diagram.

[Ans] Heart is known as the Pumping station of the Human Body in which It contains 4 Chambers, 2 Auricles and 2 Ventricles...and Have 2 Valves and 2 Septum...



- Externally Heart is having two deep grooves one is Horizontal and another is Vertical. Internally that deep groove divides the heart into 4 chambers.
- The chambers present in the ground floor called as Ventricles and those present in the first floor are called as Auricles .

- The Auricle present towards right side is called as Right Auricle and that present towards left is called as Left Auricle. Similarly The Ventricles are Right and left Ventricle.
- Both Auricles and Ventricles are separated by Auricular-Ventricular Septum. There is an opening between Auricles and Ventricles for pumping the blood are called as Valves which prevents the backward flow of blood.

2. Write the differences between -

(1)

Difference on	Mitosis	Meiosis
Division of	Body Cells	Sex Cells
Daughter cells produced	Two daughter cells are produced after mitosis	Four daughter cells are produced after meiosis
Type of Daughter Cells	Diploid	Haploid
Daughter cells are	Genetically Identical	Genetically Adverse

(2) Photosynthesis - Respiration

- Photosynthesis is energy trapping process while respiration is releasing process.
- Photosynthesis takes place in presence of sunlight on the other hand respiration is a continuous process occurring day and night.
- It builds up complex organic molecules, hence it is anabolic process where as respiration breaks down

complex organic molecules, and hence it is a catabolic process.

- Photosynthesis takes place in chloroplast. Respiration takes place in chloroplast.

OR

Explain the changes involved in the formation of seed from Ovule.

- Fusion of male and female gametes are called as fertilization. Pollen grain at stigma stimulates to give rise to pollen tubes which has 2 male nuclei. 1 Male nucleus fuses with the egg and form $2n$ zygote also known as first fertilization. The second male nucleus fuses to form triploid endosperm nucleus. Thus double fertilization occurs in embryo sac.
- Post double fertilization, due to formation of endosperm tissue by mitosis and the development of new embryo ovule size gets increased. The embryo has cotyledon and stores food.
- The zygote divides several times to form an embryo within the ovule. The ovule develops a tough coat and is converted into a seed.

3. What materials are required to prove that Oxygen is produced during Photosynthesis in the presence of light? What procedure we need to follow to perform the above experiment?

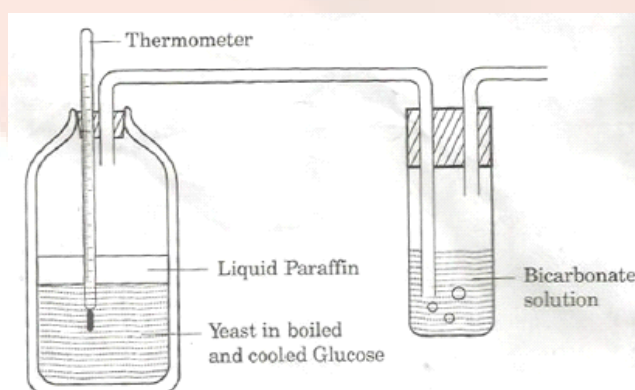
[ANS]:-

Required materials are:-

- A large Beaker
- Some Hydrilla Plant
- One Funnel
- One Test tube water a glowing splinter of wood
- A beehive arrangement etc.

Procedure for this experiment:-The Hydrilla plant should be kept in the beaker and covered with the funnel. Hydrilla plant is placed in such a way that its cutting ends remain towards the stem of the funnel. Water should be poured in the beaker in such a way that the stem of the funnel kept beneath the water level. A test tube completely filled with water should be inverted over the stem of the funnel. At this position, the experimental set should be placed in Sunlight (or under electric light in the laboratory).

OR



Observe the above diagram and answer the following questions:

(1) What does the above setting (diagram) indicate?

Answer:- Above diagram indicates the experiment, Anaerobic Respiration of Yeast.

(2) Why boiled and cooled Glucose is covered with Paraffin?

Answer:- The boiled and cooled Glucose is covered with Paraffin so that there could no aerobic respiration could occur. This Paraffin acts as a barrier between oxygen and the Glucose solution.

(3) What is the use of adding diazine green to Glucose solution? What change you notice in Glucose solution?

Answer: - To remove the oxygen from glucose we add diazene green and the changes we notice to the solution it turns in to pink colour.

(4) Why lime water is used in this experiment?

Answer:- Lime water in the presence of CO_2 or carbon dioxide turns cloudy and thus provide the reactant sugar for yeast to respire.

(5) Why bulb of thermometer is dipped in the Glucose water?

Answer:- Bulb of thermometer is dipped in the Glucose water because the bulb of a thermometer will gradually come to the thermal equilibrium with the water.

4. Read the following table and answer the questions given below :

Sl. No.	Name of the gland	Location	Hormone secreted	Response of the body to hormone
	Pituitary	Floor of brain	Somatotropin	Growth of bone.
			Gonadotropin	Activity of ovary and testis.
	Thyroid	Neck	Thyroxine	General growth rate and metabolic activity.
	Ovary	Lower abdomen	Estrogen	— Growth of the uterus and skeleton of the pelvis.
	Testis	Scrotal sac	Testosterone	Growth of male secondary sexual characters.

(1) Write the importance of glands and hormones.

[Ans] *To control sugar level and blood pressure in the body.*

(2) Which hormone is responsible for growth of bone?

[Ans] Pituitary gland.

(3) What happens if testosterone is not secreted?

[Ans] If the testosterone is not secreted erectile dysfunction happens, that is the development of

(4) Where does the gland that secretes thyroxine is located?

[ans] *neck*

(5) Which glands are common in male and female?

[ans] Pituitary glands, Thyroid.

OR

Village	Type of farmer	Income per acre on Crops			
		Paddy	Cotton	Mirchi	Maize
A	Small	7,500	9,300	5,200	5,000
	Large	26,700	38,000	16,700	12,900
B	Small	7,200	8,750	4,900	5,100
	Large	32,900	42,000	18,400	13,700 .

Observe the above table and answer the following questions.

(1) Which crop is most suitable to cultivate for small farmer in both the villages?

[ans] cotton

(2) If you are a large farmer, which crop do you select to cultivate?

[ans] cotton

(3) What similarities you have identified in village A and village B?

[ans] small farmers are earning less in comparison of large farmer.

(4) Which is the lowest income crop?

[ans] Maize

(5) Is there any relationship between production of crops and income? How?

[ans] The increase in the quality and quantity leads to the increase in the income of the farmer.

SECTION - II

(6x2=12 marks)

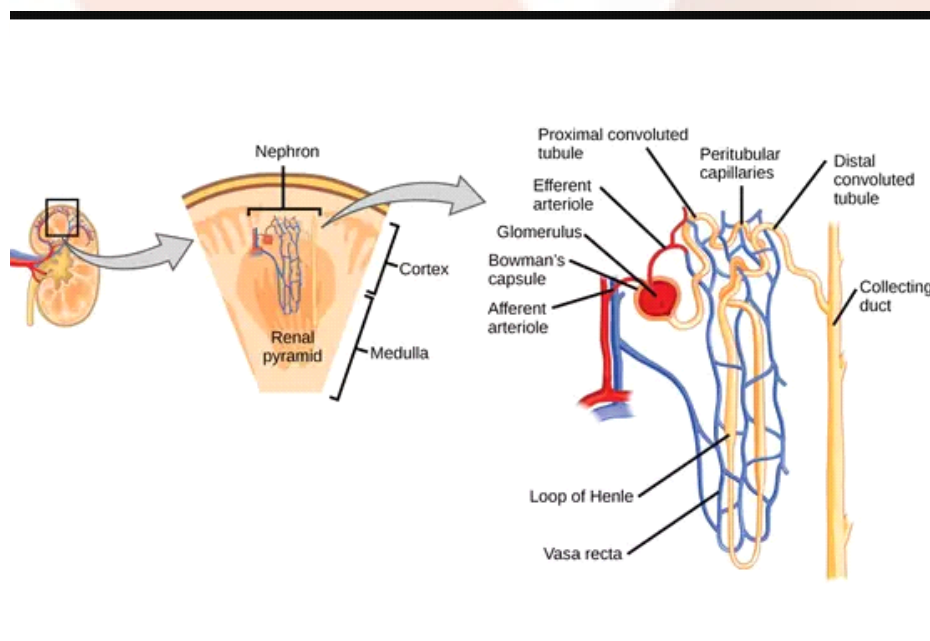
Note:

- (i) Answer all the questions.
- (ii) Each question carries TWO marks.
- (iii) Answer each question in 4 to 5 sentences.

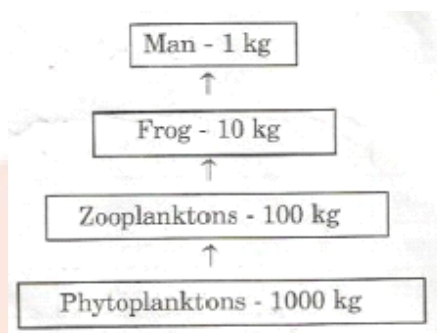
5. "Forest is a renewable resource." Do you agree? Justify.

[ans] Yes, forest are the renewable resources because unlike non-renewable resources the forest can grow in short period of time and can be used repeatedly as fuels and other resources full filling the conditions of renewable resources, thus fell in this category.

6. Draw the structure of an excretory organ, which contains Bowman's capsule and loop of Henle and label it.



7. Explain the Flow-chart given below :



[ans] This flow chart shows the trophic efficiency of organism which prove that the efficiency decreases by 10% as we move upwards. Starting from phytoplankton – zooplanktons – frog – man varies in the sequence 1000kg-100kg-10kg-1kg respectively.

8. What questions do you pose to your teacher to understand "blood clotting"?

[ans]

Q. What prevents clotting of blood in blood vessels?

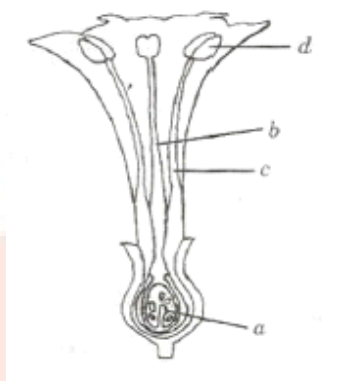
Q. The ions responsible for the process of blood clot

9. What happens if Meiosis does not take place in reproductive cells?

[ans] Meiosis doesn't occur will lead to doubling of chromosomes which results in abnormalities.

If meiosis doesn't occur, fusion of gametes will result in a doubling of the chromosomes for each successive reproduced generation

10. Identify the flower parts a, b, c, d and write their main function.



[Ans]- a- ovule

b- Style

c- Filament

d- Anther.

- Style – Holds up the stigma
- Anthers – Pollen producing part
- Filaments – They hold up the anthers

SECTION - III

(7x1=7marks)

Note:

(0) Answer all the questions.

(ii) Each question carries ONE mark.

(iii) Answer each question in 1 or 2 sentences.

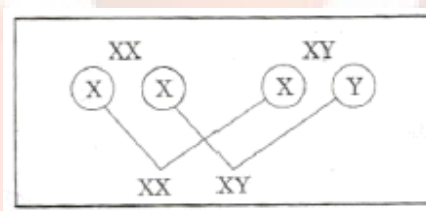
11. To create awareness on “Water conservation” in your locality, what slogan you will suggest?

[Ans] - To create awareness about "water conservation" in our area, my slogan would be "Water is life"

12. Why stomach is structured like a bag rather than like a tube?

- The food taken has to remain in the stomach for proper digestion.
- If it is like a tube it will just pass down without undergoing changes and cannot remain in the stomach for long time.

13. Observe the following Flow-chart and answer the question.



Who decides the sex of the baby - mother or father? How?

[Ans] - Females have two 'X' chromosomes in their cells (XX). Males have one 'X' and one 'Y' chromosomes in their cells (XY). Men determine the sex of a baby depending on whether their sperm is carrying an X or Y chromosome. An X chromosome combines with the mother's X chromosome to make a baby girl (XX) and a Y chromosome will combine with the mother's to make a boy (XY).

14. Why KOH is used in Moll's half leaf experiment?

[Ans] - KOH is used in Moll's half leaf experiment because it absorbs carbon dioxide

15. A plant which grows near a window bends towards Sunlight. Write the reason for it.

[Ans] - Plants grow towards the sunlight due to photo tropism {Plants grow towards the sunlight, Such type of response of a plant to light is called photo tropism (photo means light, tropism means movement)}

16. Prepare your own tabular column to get information about food deficiency diseases from a doctor.

Vitamin/ Mineral	Deficiency disease/disorder	Symptoms
Vitamin A	Loss of vision	Poor vision, loss of vision in darkness (night), sometimes complete loss of vision
Vitamin B1	Beriberi	Weak muscles and very little energy to work
Vitamin C	Scurvy	Bleeding gums, wounds take longer time to heal
Vitamin D	Rickets	Bones become soft and bent
Calcium	Bone and tooth decay	Weak bones, tooth decay
Iodine	Goiter	Glands in the neck appear swollen, mental disability in children
Iron	Anaemia	Weakness

17. What examples you will give to prove that Lamarckism is not correct?

[Ans] - August Weismann tested this theory by an experiment on rats. He removed tails of parental rats. He observed that their offspring's are normal with tails. He has done it again and again for twenty two generations but still offspring are

normal with tails. He proved that the bodily changes are not inherited. So they won't be passed to its offspring

SECTION – IV

($10 \times \frac{1}{2} = 5$ marks)

Note:

- (i) Answer all the questions.
- (ii) Each question has FOUR choices. Choose the correct answer for each question and write the relevant alphabet [A, B, C, D] against the question number in your answer booklet.
- (iii) Each question carries mark.

18. Deficiency of Folic acid causes

- (A) Anaemia
- (B) Pellagra
- (C) Glossitis
- (D) Rickets

[Ans] A - Deficiency of Folic acid causes Anaemia

19. Acid present in Gastric juice

- (A) Sulphuric acid
- (B) Hydrochloric acid
- (C) Nitrous acid
- (P) Phosphoric acid

[Ans] B - Pure gastric juice contains large amounts of HCl

20. A person has lost control on emotions. Which part of brain is responsible for it?

- (A) Cerebrum
- (B) Diencephalon
- (C) Mid brain
- (D) Cerebellum

[Ans] A - The cerebrum is responsible for lost control over emotions.

21. Identify the animal in which excretory organ is absent.

- (A) Bird
- (B) Amoeba
- (C) Sponge
- (D) Band C

[Ans] C - Sponges and coelenterates do not have specific excretory organs as water bathes almost all their cells.

22. Which of the following practice is suitable to farmer with less water?

Resource?

- (i) Select short term crops.
- (ii) Cultivate commercial crops.
- (iii) Adapt drip irrigation system.
- (iv) Crop holiday.

- (A) (i), (ii)
- (B) (i), (ii), (iii)
- (C) (i), (iv)
- (D) (ii), (iv)

[Ans]

23. Which plant hormone is responsible for closing of Stomata?

- (A) Absciscic acid
- (B) Auxin
- (C) Cytokinin
- (D) Ethylene

[Ans] Absciscic acid is responsible for closing stomata.

